

SHANNON, VA, USA

WMO: 724033

Lat: 38.267N

Lon: 77.449W

Elev: 85

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03706

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	18.7	-0.3	5.4	20.6	4.0	6.8	23.9	23.0	45.4	20.3	43.8	3.5	330	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	95.3	77.9	92.8	76.6	90.5	75.7	80.4	90.3	79.0	88.6	77.7	86.6	6.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	147.0	84.1	76.7	139.6	83.0	75.1	132.3	81.8	44.0	89.6	42.5	88.3	41.1	87.1	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.3	13.5	6.5	98.9	6.3	2.8	1.9	100.9	-1.7	102.6	-5.3	104.2	-9.8	106.2
			5.8	82.1	6.1	1.9	1.4	83.5	-2.1	84.6	-5.6	85.7	-10.0	87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	37.1	39.7	47.1	57.6	66.6	74.7	78.8	77.6	70.6	59.2	48.3	40.8
	DBStd	16.85	10.78	9.80	9.95	8.98	7.82	6.04	4.90	4.93	6.54	8.28	8.59	9.01
	HDD50	1400	424	312	178	27	0	0	0	0	14	132	313	
	HDD65	3974	865	708	560	258	78	7	0	24	219	504	751	
	CDD50	4419	24	25	88	254	515	740	893	855	618	300	80	27
	CDD65	1521	1	1	6	36	128	296	428	390	192	40	2	1
	CDH74	14763	3	5	82	468	1274	2852	4377	3739	1578	354	26	5
	CDH80	6179	0	0	15	150	448	1192	2050	1671	559	92	2	0
	WSAvg	4.9	5.6	5.7	6.4	6.2	5.0	4.5	3.8	3.5	4.0	4.3	4.7	4.7
	PrecAvg	41.2	3.2	2.9	3.8	3.0	3.9	3.3	4.1	3.5	3.8	3.5	3.3	3.3
(15) Precipitation	PrecMax	59.7	7.6	6.4	9.6	7.0	8.6	10.8	11.7	10.0	10.9	8.0	7.2	7.1
	PrecMin	27.8	0.4	0.3	1.0	0.4	0.6	0.7	0.8	0.5	0.5	0.2	0.2	0.7
	PrecStd	8.1	1.5	1.6	1.9	1.4	1.9	2.0	2.6	2.3	2.5	2.0	1.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	73.3	81.8	89.9	91.5	96.6	99.5	98.8	93.6	87.6	78.1	72.9
		MCWB	61.0	60.5	64.1	69.2	73.8	78.2	79.1	77.8	75.4	73.1	64.3	61.7
	2%	DB	65.7	66.4	75.2	84.2	88.7	92.6	95.6	94.8	90.2	82.1	72.4	65.9
		MCWB	59.5	56.3	60.0	66.0	72.7	76.4	78.4	77.8	74.7	69.3	61.3	59.9
	5%	DB	61.1	61.4	70.3	79.5	85.4	90.2	92.9	91.4	86.3	77.7	68.0	61.0
		MCWB	54.3	52.0	57.8	64.0	71.1	75.6	77.1	76.6	72.8	67.2	59.1	54.5
	10%	DB	54.3	55.6	64.8	74.6	81.6	87.6	90.4	89.1	82.3	73.3	63.7	55.3
		MCWB	47.8	47.5	54.2	61.2	68.9	74.3	76.4	75.4	70.9	65.2	56.3	49.4
	0.4%	WB	64.7	63.5	65.9	72.8	77.7	81.2	83.1	82.2	78.7	75.8	67.0	66.0
		MCDB	67.7	70.5	77.4	83.3	86.7	91.3	93.2	91.7	86.5	81.6	73.3	69.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	60.0	58.8	62.7	68.8	75.1	78.8	80.8	80.1	76.9	73.0	64.3	61.0
		MCDB	65.5	64.6	72.1	80.2	84.3	88.0	90.5	90.2	85.3	78.7	69.3	65.2
	5%	WB	54.8	53.2	59.6	65.9	73.1	77.2	79.3	78.5	75.3	70.0	61.3	55.7
		MCDB	60.4	58.5	68.2	76.2	81.6	86.8	89.3	87.7	82.5	75.9	66.8	59.8
	10%	WB	48.4	48.5	55.8	63.1	71.1	75.7	77.9	77.2	73.5	66.7	57.2	50.4
		MCDB	53.2	55.5	63.7	72.0	79.1	84.1	87.0	85.2	79.0	71.8	62.3	54.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	20.3	21.8	23.4	21.7	20.4	20.0	20.2	20.4	22.0	22.1	19.0
		MCDBR	25.9	27.5	29.5	29.7	26.2	23.6	23.1	23.5	24.6	26.6	28.0	24.9
		MCWBR	19.8	18.9	17.6	14.6	12.5	10.4	8.6	9.1	10.6	14.8	17.8	18.7
	5% WB	MCDBR	23.9	24.1	25.4	25.3	22.1	21.0	20.8	20.6	19.5	21.4	21.9	21.7
		MCWBR	20.1	18.7	16.8	13.9	11.3	9.9	8.5	9.0	9.7	13.0	16.5	18.3
(40) Clear-Sky Solar Irradiance	taub	0.304	0.318	0.354	0.395	0.434	0.488	0.508	0.509	0.416	0.358	0.323	0.308	
	taud	2.509	2.451	2.403	2.304	2.224	2.113	2.082	2.059	2.300	2.464	2.519	2.534	
	Ebn at Noon	280	289	288	280	270	255	248	244	263	271	270	270	
	Edh at Noon	25	30	35	40	45	50	51	51	38	29	24	23	
(44) All-Sky Solar Radiation	RadAvg	693	948	1260	1573	1782	1954	1894	1699	1390	1065	784	611	
	RadStd	61	105	91	117	181	144	97	83	159	121	64	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.89	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+248	+151

Nomenclature: See separate page